



- 1 A manager is investigating the times taken by employees to complete a particular task as a result of the introduction of new technology. He claims that the mean time taken to complete the task is reduced by more than 0.4 minutes. He chooses a random sample of 10 employees. The times taken, in minutes, before and after the introduction of the new technology are recorded in the table.

| Employee                   | A    | B   | C    | D    | E    | F    | G    | H    | I    | J    |
|----------------------------|------|-----|------|------|------|------|------|------|------|------|
| Time before new technology | 10.2 | 9.8 | 12.4 | 11.6 | 10.8 | 11.2 | 14.6 | 10.6 | 12.3 | 11.0 |
| Time after new technology  | 9.6  | 8.5 | 12.4 | 10.9 | 10.2 | 10.6 | 12.8 | 10.8 | 12.5 | 10.6 |

- (a) Test at the 10% significance level whether the manager's claim is justified. [7]

$H_0: \mu_d = 0.4$ ,  $H_1: \mu_d > 0.4$  where  $\mu_d = \mu_{\text{before}} - \mu_{\text{after}}$  of population

It is a one-tailed test.  $\bar{D} = \text{Time}_{\text{before}} - \text{Time}_{\text{after}} \sim N\left(\mu_d, \frac{s_d^2}{10}\right)$

|           | A   | B   | C | D   | E   | F   | G   | H    | I    | J   |
|-----------|-----|-----|---|-----|-----|-----|-----|------|------|-----|
| $\bar{D}$ | 0.6 | 1.3 | 0 | 0.7 | 0.6 | 0.6 | 1.8 | -0.2 | -0.2 | 0.4 |

$$\sum d = 5.6, \sum d^2 = 0.6^2 + 1.3^2 + \dots + 0.2^2 + 0.4^2 = 6.74, n = 10$$

From this,

$$\bar{d} = \frac{\sum d}{n} = 0.56, s_d^2 = \frac{1}{n-1} \left( \sum d^2 - \frac{(\sum d)^2}{n} \right) = \frac{901}{2250} = 0.40044$$

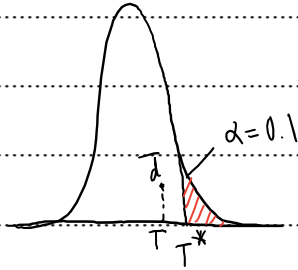
Since  $n < 30$ , we cannot use CLT and must use t-test.

So,

$$\text{test statistic } T = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}} = \frac{0.56 - 0.4}{\sqrt{\frac{0.40044}{10}}} = 0.7996$$

We need critical value  $T^* = t_{(1-\alpha), n-1} = t_{0.9, 9} = 1.383$

We can now plot the sampling distribution of  $\bar{D}$



Clearly,  $T$  is not in the rejection/critical region.

$0.7996 < 1.383$ , so we do not reject  $H_0$ .

There is insufficient evidence to suggest that the mean time to complete the task has reduced by more than 0.4 minutes.

OR

There is insufficient evidence to suggest that the manager's claim is justified.

We did not use a Wilcoxon matched-pairs signed rank test here because it was explicitly about the mean. If the manager claimed about the average, we could have used either test.

(b) State an assumption that is necessary for this test to be valid.

[1]

Underlying distribution of population (of differences) is normal.

- 2 The probability generating function,  $G_Y(t)$ , of the random variable  $Y$  is given by

$$G_Y(t) = 0.04 + 0.2t + 0.37t^2 + 0.3t^3 + 0.09t^4.$$

- (a) Find  $\text{Var}(Y)$ .

[4]

$$\text{Var}(Y) = G''_Y(1) + G'_Y(1) - [G'_Y(1)]^2$$

$$G'_Y(t) = 0.2 + 0.74t + 0.7t^2 + 0.36t^3$$

$$G''_Y(t) = 0.74 + 1.4t + 1.08t^2$$

$$\text{So } G'_Y(1) = 2.2, G''_Y(1) = 3.62$$

Substitute them into first equation,

$$\text{Var}(Y) = 3.62 + 2.2 - 2.2^2 = 0.98$$

The random variable  $Y$  is the sum of two independent observations of the random variable  $X$ .

- (b) Find the probability generating function of  $X$ , giving your answer as a polynomial in  $t$ . [3]

By the given condition,

$$G_Y(t) = G_{X_1+X_2}(t) = G_{X_1}(t) \cdot G_{X_2}(t) = [G_X(t)]^2$$

★ Since we are told that  $G_X(t)$  is a polynomial, let it be

$$G_X(t) = a + bt + ct^2$$

and we have no more higher degree terms, since highest degree in  $G_Y(t)$  is 4. Therefore,

$$G_Y(t) = (a + bt + ct^2)^2 = a^2 + b^2t^2 + c^2t^4 + 2abt + 2act^2 + 2bct^3$$

Matching the coefficients to the known eq<sup>n</sup> of  $G_Y(t)$ ,

$$a^2 = 0.04, a = 0.2$$

$$2ab = 0.2, b = \frac{0.1}{a} = \frac{0.1}{0.2} = 0.5$$

$$c^2 = 0.09, c = 0.3$$

$$\text{Finally, } G_X(t) = 0.2 + 0.5t + 0.3t^2$$

We did not consider  $a = -0.2$  and  $c = -0.3$  since  $a, b, c$  are probabilities. Once we found any two of  $a, b, c$ , we could have alternatively found the third by subtracting their total from 1.

- 3 The continuous random variable  $X$  has probability density function  $f$  given by

$$f(x) = \begin{cases} kx(4-x) & 0 \leq x < 2, \\ k(6-x) & 2 \leq x \leq 6, \\ 0 & \text{otherwise,} \end{cases}$$

where  $k$  is a constant.

- (a) Show that  $k = \frac{3}{40}$ . [1]

$$k \int_0^2 (4x - x^2) dx + k \int_2^6 (6 - x) dx = 1, \quad k \left[ 2x^2 - \frac{1}{3}x^3 \right]_0^2 + k \left[ 6x - \frac{1}{2}x^2 \right]_2^6 = 1$$

$$k \left( \frac{16}{3} \right) + k(8) = 1, \quad \frac{40}{3}k = 1, \text{ so } k = \frac{3}{40}.$$

- (b) Given that  $E(X) = 2.5$ , find  $\text{Var}(X)$ . [3]

$$\text{Var}(X) = E(X^2) - 2.5^2$$

$$E(X^2) = \int_0^2 x^2 \cdot kx(4-x) dx + \int_2^6 x^2 \cdot k(6-x) dx$$

$$= k \int_0^2 (4x^3 - x^4) dx + k \int_2^6 (6x^2 - x^3) dx$$

$$= k \left[ x^4 - \frac{1}{5}x^5 \right]_0^2 + k \left[ 2x^3 - \frac{1}{4}x^4 \right]_2^6$$

$$= \frac{48}{5}k + 96k = 7.92$$

Thus,

$$\text{Var}(X) = 7.92 - 2.5^2 = 1.67$$

(c) Find the median value of  $X$ .

[4]

From (a), we know that  $f(x)$  in  $0 \leq x < 2$  contributes  $\frac{16}{3}k = 0.4$  total probability density, so median must lie in  $2 \leq x \leq 6$ .

Therefore,

$$0.4 + \int_2^m k(6-x) dx = 0.5,$$

$$\int_2^m (6-x) dx = \frac{0.5 - 0.4}{k} = \frac{4}{3}$$

$$\left[ 6x - \frac{1}{2}x^2 \right]_2^m = \frac{4}{3}$$

$$6m - 0.5m^2 - 10 - 4/3 = 0$$

$$3m^2 - 36m + 68 = 0$$

$$m = \frac{36 \pm \sqrt{480}}{6} \approx 2.35, 9.65$$

out of domain

So  $m = 2.35$  (3 sf) only.

- 4 A scientist is investigating the numbers of a particular type of butterfly in a certain region. He claims that the numbers of these butterflies found per square metre can be modelled by a Poisson distribution with mean 2.5. He takes a random sample of 120 areas, each of one square metre, and counts the number of these butterflies in each of these areas. The following table shows the observed frequencies together with some of the expected frequencies using the scientist's Poisson distribution.

| Number per square metre | 0    | 1     | 2     | 3     | 4   | 5    | 6    | $\geq 7$ |
|-------------------------|------|-------|-------|-------|-----|------|------|----------|
| Observed frequency      | 12   | 20    | 36    | 32    | 13  | 6    | 1    | 0        |
| Expected frequency      | 9.85 | 24.63 | 30.78 | 25.65 | $p$ | 8.02 | 3.34 | $q$      |

- (a) Find the values of  $p$  and  $q$ , correct to 2 decimal places. [2]

Total no. of observations  $N = 120$

$$E_r = N \times \frac{e^{-2.5} \cdot 2.5^r}{r!}, \text{ so } p = E_4 = 16.03 \text{ (2 dp)}$$

$$q = N - \sum_{i=0}^6 E_i = 120 - (9.85 + 24.63 + \dots + 3.34) = 1.70 \text{ (2 dp)}$$

$$p = 16.03$$

$$q = 1.70$$

- (b) Carry out a goodness of fit test, at the 10% significance level, to test the scientist's claim. [6]

We must do a  $\chi^2$ -test.

We reformat the table so each expected frequency  $E_i \geq 5$ .

| No. per $m^2$ | $O_i$ | $E_i$ | $(O_i - E_i)^2 / E_i$ |
|---------------|-------|-------|-----------------------|
| 0             | 12    | 9.85  | 0.4693                |
| 1             | 20    | 24.63 | 0.8704                |
| 2             | 36    | 30.78 | 0.8853                |
| 3             | 32    | 25.65 | 1.5720                |
| 4             | 13    | 16.03 | 0.5727                |
| 5             | 6     | 8.02  | 0.5088                |
| $\geq 6$      | 1     | 5.04  | 3.2384                |

(always round this column to 4 dp so you are safe against the 3 dp table critical value)

$H_0$ :  $Po(2.8)$  is a good fit for the data

$H_1$ :  $Po(2.8)$  is not a good fit for the data

$n$  = no. of rows = 7

$$\text{Test statistic } \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 0.4693 + 0.8704 + \dots + 3.2384 = 8.1169$$

Degrees of freedom  $\nu = n - 1 = 6$

$$\text{Critical value } \chi_{\nu}^{2*}(1-\alpha) = \chi_6^{2*}(0.90) = 10.64$$

Comparing  $\chi^2$  and  $\chi^{2*}$ ,

$8.1169 < 10.64$ , so we do not reject  $H_0$ .

There is insufficient evidence to suggest that  $Po(2.8)$  is not a good fit.

$\nu = n - 1$  only because we estimated no extra parameters besides the last  $E_i$ .

- 5 Raman is researching the heights of male giraffes in a particular region. Raman assumes that the heights of male giraffes in this region are normally distributed. He takes a random sample of 8 male giraffes from the region and measures the height, in metres, of each giraffe. These heights are as follows.

5.2    5.8    4.9    6.1    5.5    5.9    5.4    5.6

- (a) Find a 90% confidence interval for the population mean height of male giraffes in this region. [5]

$$\text{Confidence Interval (CI)} = \bar{x} \pm t_{(1-\frac{\alpha}{2}), n-1} \cdot \frac{s}{\sqrt{n}}$$

$$n=8, \sum x = 44.4, \sum x^2 = 247.48$$

$$\bar{x} = \frac{\sum x}{n} = 5.55, s^2 = \frac{1}{7} \left( 247.48 - \frac{44.4^2}{8} \right) = \frac{53}{350}$$

$$t_{0.95, 7} = 1.895$$

$$CI = 5.55 \pm 1.895 \times \sqrt{\frac{53/350}{8}} = 5.55 \pm 0.2607$$

Therefore,

$$95\% \text{ confidence interval} = (5.29, 5.81)$$

We rounded the final answer to 3 sf.

Raman claims that the population mean height of male giraffes in the region is less than 5.9 metres.

- (b) Test at the 2.5% significance level whether this sample provides sufficient evidence to support Raman's claim. [4]

$$H_0: \mu = 5.9$$

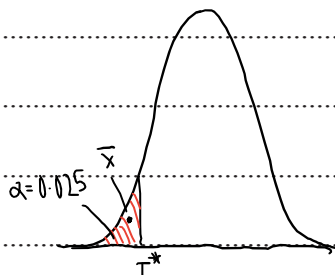
$$H_1: \mu < 5.9$$

It is a single sample t-test,  
and one tailed.

$$\text{Test statistic } T = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{5.55 - 5.9}{0.1376} = -2.544$$

$$\text{Critical value } T^* = -t_{(1-\alpha), n-1} = -t_{0.975, 7} = -2.365$$

Plotting the sampling distribution,



Clearly,  $T$  is in the critical/rejection region.

Explicitly comparing  $T$  and  $T^*$ ,

$$-2.544 < -2.365, \text{ so we reject } H_0.$$

There is sufficient evidence to suggest that mean height of the giraffes population is less than 5.9 metres.

OR

There is sufficient evidence to support Raman's claim.

- 6 A teacher at a large college gave a mathematical puzzle to all the students. The median time taken by a random sample of 24 students to complete the puzzle was 18.0 minutes. The students were then given practice in solving puzzles. Two weeks later, the students were given another mathematical puzzle of the same type as the first. The times, in minutes, taken by the random sample of 24 students to complete this puzzle are as follows.

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 18.2 | 17.5 | 16.4 | 15.1 | 20.5 | 26.5 | 19.2 | 23.2 |
| 17.9 | 18.8 | 25.8 | 19.9 | 17.7 | 16.2 | 17.3 | 16.6 |
| 17.1 | 20.1 | 20.3 | 12.6 | 16.0 | 21.4 | 22.7 | 18.4 |

The teacher claims that the practice has not made any difference to the average time taken to complete a puzzle of this type.

Carry out a Wilcoxon signed-rank test, at the 10% significance level, to test whether there is sufficient evidence to reject the teacher's claim. [10]

$H_0: m = 18.0$ ,  $H_1: m \neq 18.0$  where  $m$  is the median of the population.

Let  $R$  represent the signed rank.

| Time | Time - m | R   | Time | Time - m | R   | Time | Time - m | R   |
|------|----------|-----|------|----------|-----|------|----------|-----|
| 18.2 | 0.2      | 2   | 17.9 | -0.1     | -1  | 17.1 | -0.9     | -8  |
| 17.5 | -0.5     | -5  | 18.8 | 0.8      | 7   | 20.1 | 2.1      | 15  |
| 16.4 | -1.6     | -11 | 25.8 | 7.8      | 23  | 20.3 | 2.3      | 16  |
| 15.1 | -2.9     | -18 | 19.9 | 1.9      | 13  | 12.6 | -5.4     | -22 |
| 20.5 | 2.5      | 17  | 17.7 | -0.3     | -3  | 16.0 | -2.0     | -14 |
| 26.5 | 8.5      | 24  | 16.2 | -1.8     | -12 | 21.4 | 3.4      | 19  |
| 19.2 | 1.2      | 9   | 17.3 | -0.7     | -6  | 22.7 | 4.7      | 20  |
| 23.2 | 5.2      | 21  | 16.6 | -1.4     | -10 | 18.4 | 0.4      | 4   |

Sum of positive ranks  $P = 190$

Sum of negative ranks  $Q = 110$

Hence, test statistic  $T = \min(P, Q) = 110$

For critical value  $T^*$ , since  $n > 20$ , we use a normal approximation on  $Q$ .

$$E(Q) = \frac{1}{4}n(n+1), \text{Var}(Q) = \frac{1}{24}n(n+1)(2n+1)$$

$$= 150$$

$$= 1225$$

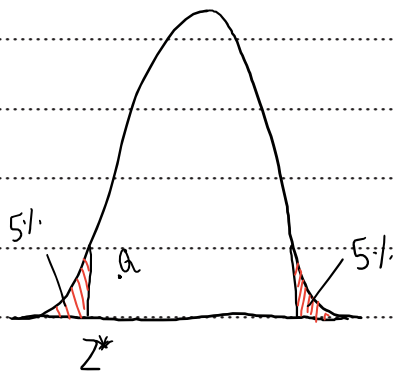
$$\text{So } Q \sim N(150, 1225)$$

$$\text{Our new test statistic } Z = \frac{Q + 0.5 - 150}{\sqrt{1225}} = \frac{110 - 149.5}{35} = -1.129$$

$$\text{Our new critical value } Z^* = -\Phi^{-1}\left(1 - \frac{\alpha}{2}\right) = -\Phi^{-1}(0.95) = -1.645$$

because two-tailed

Plot our normal curve,



Clearly,  $Q$  is not in the critical/rejection region.

Comparing  $Z$  and  $Z^*$  explicitly,

$$-1.129 > -1.645, \text{ so we do not reject } H_0.$$

There is insufficient evidence to reject the teacher's claim.

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.



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